UNIT 3 – BINARY SEARCH TREE

TOPIC: MENU-DRIVEN BINARY SEARCH TREE OPERATIONS

1. Introduction

A Binary Search Tree (BST) is a non-linear data structure made up of nodes connected through links. For each node, values smaller than the node are stored in its left subtree, and values greater than the node are stored in its right subtree. This project demonstrates common BST operations using a menu-driven C++ program.

2. Problem Statement

Write a menu-driven C++ program to perform insertion, searching, deletion, inorder traversal, preorder traversal, and postorder traversal on a Binary Search Tree (BST).

3. Objectives

1. To create and manage a Binary Search Tree.

2. To insert a node at its correct position in the BST.

3. To search for a specified value in the BST.

4. To delete a node while maintaining BST properties.

5. To perform inorder, preorder, and postorder traversals.

6. To understand tree-based data structures and their operations.

4. Data Structure Used

Binary Search Tree (BST): Each node contains a value and links to its left and right child. The left subtree contains smaller values, while the right subtree contains larger values. This structure supports searching, insertion, deletion, and traversal operations.

5. Algorithm

Step 1: Start the program.

Step 2: Create an empty Binary Search Tree (BST).

Step 3: Display the menu with the following operations:
1. Insert a node
2. Search for a node
3. Delete a node
4. Inorder traversal
5. Preorder traversal
6. Postorder traversal
7. Exit

Step 4: Accept the user's choice.

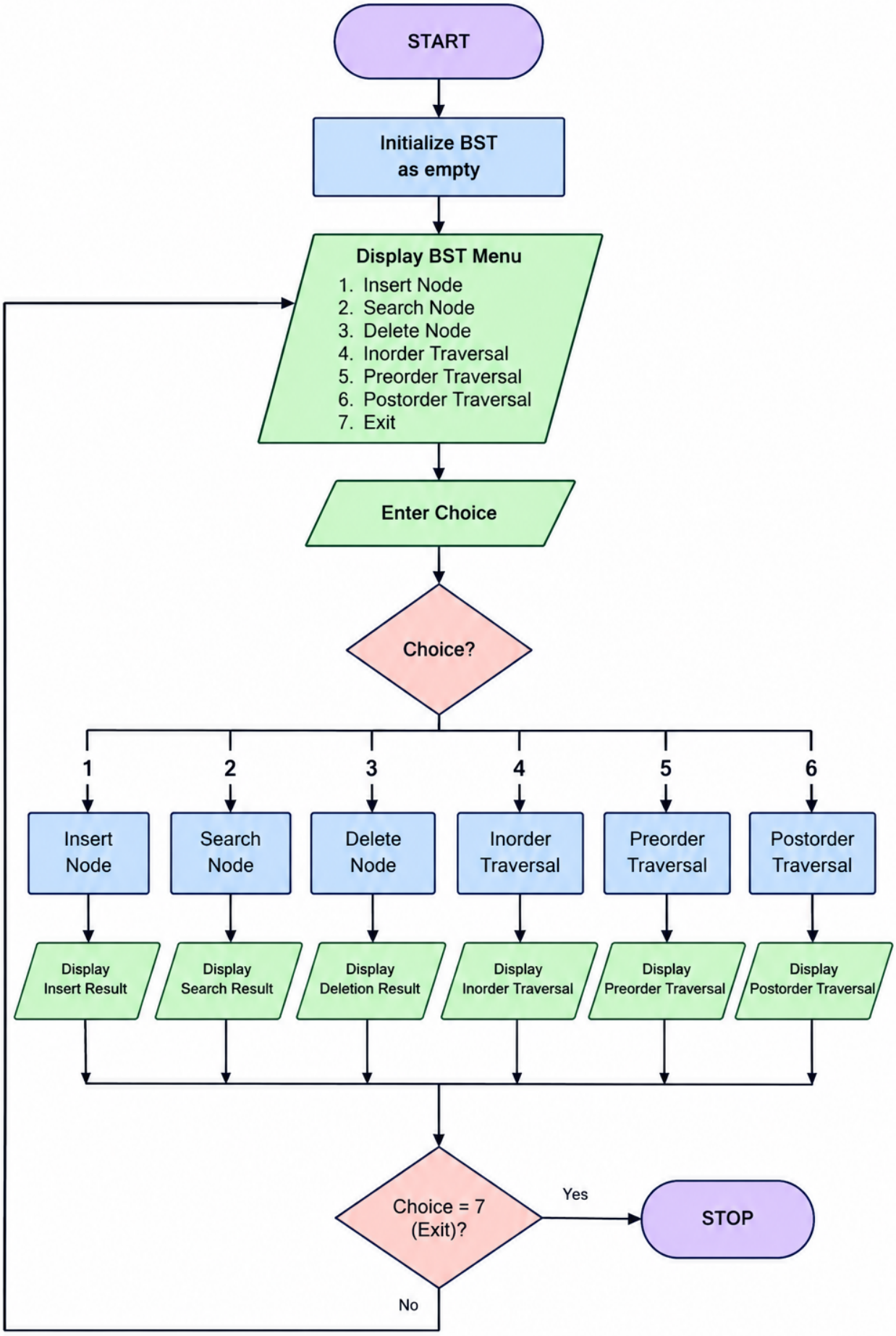
Step 5: Perform the selected operation:
• Insertion: Insert the given value into its correct position in the BST.
• Searching: Search for the given value and display whether it exists.
• Deletion: Delete the specified value while maintaining BST properties.
• Inorder Traversal: Display nodes in left-root-right order.

• Preorder Traversal: Display nodes in root-left-right order.
• Postorder Traversal: Display nodes in left-right-root order.

Step 6: Repeat the menu until the user selects Exit.

Step 7: Stop the program.

6. Flowchart


START
Initialize BST as empty
Display BST Menu
1. Insert Node
2. Search Node
3. Delete Node
4. Inorder Traversal
5. Preorder Traversal
6. Postorder Traversal
7. Exit
Enter Choice
Choice?
1
2
3
4
5
6
Insert Node
Search Node
Delete Node
Inorder Traversal
Preorder Traversal
Postorder Traversal
Display Insert Result
Display Search Result
Display Deletion Result
Display Inorder Traversal
Display Preorder Traversal
Display Postorder Traversal
Choice = 7 (Exit)?
Yes
STOP
No

7. Working of the Project

The program displays a menu and accepts the user's choice. Based on the selected option, it inserts a value, searches for a value, deletes a node, or displays the BST using inorder, preorder, or postorder traversal. The menu is repeated until the user selects Exit.

8. Programming Language and Concepts

• Programming Language: C++

• Structures and pointers

• Dynamic memory allocation

• Binary Search Tree

• Insertion, searching, and deletion

• Inorder, preorder, and postorder traversal

9. Applications

• Maintaining and retrieving sorted data.

• Searching for values in a collection.

• Implementing lookup systems and dictionaries.

• Learning and practising tree-based data structures.

10. Advantages

1. Supports searching, insertion, and deletion operations.

2. Inorder traversal displays values in sorted order.

3. Provides different methods to traverse nodes.

4. Helps in understanding non-linear data structures.

5. A menu-driven interface makes the program easy to operate.

11. Conclusion

Thus, the menu-driven C++ program demonstrates insertion, searching, deletion, and inorder, preorder, and postorder traversal operations on a Binary Search Tree. It helps in understanding BST properties and basic tree operations.